

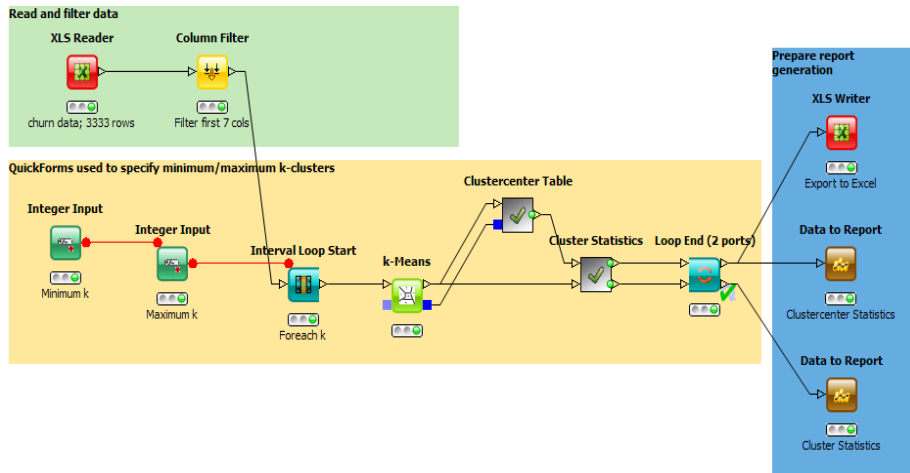
KNIME Spring Summit
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Churn / Cancelation / Retention Analysis

This workflow can be found on the KNIME Workflow Public Server under
050_Applications/050003_ChurnAnalysis

This KNIME workflow focuses on identifying classes of telecommunication customers that churn using K-Means. As in all exploratory data mining, it is unknown beforehand what number of clusters will be appropriate, therefore the workflow allows the user to specify different numbers of clusters for K-means to calculate. Each set of clusters has its' cluster centers and variances combined and presented both as a report and automatically into a spreadsheet so that clusters can be interpreted.



This workflow uses QuickForms and flow variables to provide runtime flexibility, a metanode that can be reused for combining the cluster information, report generation as well as integration with Microsoft Office products.

Churn Analysis with k-Mean Clustering										
Clustername	Clustersize	Account Length	VMail Message	Day Mins	Eve Mins	Night Mins	Intl Mins	CustServ Calls	k	
cluster_2	1133	101	8	228	219	225	10	2	3	
cluster_1	1208	100	8	129	211	214	10	2	3	
cluster_2	992	102	8	190	169	157	10	2	3	
cluster_2	850	100	8	229	237	212	10	2	4	
cluster_1	890	101	8	123	224	212	10	2	4	
cluster_2	761	102	8	185	189	140	10	2	4	
cluster_3	832	102	8	186	150	233	10	2	4	
cluster_3	645	90	9	232	242	211	10	2	5	
cluster_1	707	107	8	142	236	235	10	2	5	
cluster_2	662	116	8	212	191	149	10	2	5	
cluster_3	666	100	8	195	148	238	10	2	5	
cluster_4	653	91	8	121	187	169	10	2	5	
cluster_3	510	100	8	240	209	244	10	2	6	
cluster_1	633	107	8	141	220	247	10	2	6	
cluster_2	579	116	8	220	182	153	10	2	6	
cluster_3	525	89	8	176	137	221	10	2	6	
cluster_4	515	101	7	114	192	164	10	2	6	
cluster_5	567	91	9	190	261	176	10	2	6	

The churn data is available as a free download from Ian Pardue via <http://ebookbrowse.com/churn-xls-d139249372> and www.iainpardoe.com/teaching/dsc433/data/Churn.xls